The Disruption of Knowledge Ecosystems by Generative AI: Evidence from Stack Overflow, GitHub, and Stack Exchange Communities

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Submitted to ACM CSCW 2026
March 2026

Abstract

We present the first large-scale longitudinal study examining how generative AI reshapes knowledge ecosystems. Using eight years of data (January 2018–December 2025) from Stack Overflow (SO), GitHub, and 22 Stack Exchange communities, we document a striking "scissors effect": SO programming question activity collapsed by 98.1%, while GitHub repository creation surged 446.2%. A difference-in-differences (DID) model with 20 control variables confirms that ChatGPT's release causally accelerated this divergence (β_1=−2.26***, β_2=+3.82***, R^2=0.989, N=2,390).

Beyond volume, we uncover four structural transformations through LLM-based classification of 128,219 questions: (1) a quality dilution paradox—question length increased +17.8% while answer rates fell −23.2%; (2) the Debug question collapse predates ChatGPT by three years (−20pp in 2018–2019); (3) Conceptual questions exceeded How-to questions for the first time in 2024; (4) Philosophy SE is the only community showing growth (+54.6%), as AI itself becomes a philosophical subject. Critically, AI Replaceability Index (ARI) shows no significant correlation with activity decline (r=0.23, n.s.), suggesting behavioral substitution rather than content substitution as the core mechanism.

CCS Concepts: • Human-centered computing → Collaborative and social computing; Computer supported cooperative work; Social content sharing; • Computing methodologies → Natural language processing.

Keywords: generative AI; Stack Overflow; GitHub; knowledge community; difference-in-differences; ChatGPT; knowledge behavior; platform disruption

1. Introduction

1.1 Motivation

The release of ChatGPT on November 30, 2022 marked a turning point in how people access, produce, and share technical knowledge. Unlike prior automation waves, generative AI directly challenges knowledge work itself: searching for information, understanding concepts, debugging code, and synthesizing answers. How this capability reshapes knowledge communities is one of the most consequential empirical questions in computer-supported cooperative work.

Online knowledge communities—particularly Stack Overflow (SO) and GitHub—have served as critical infrastructure for open knowledge production since 2008. Stack Overflow accumulated over 23 million questions by 2024 and was considered the most important learning resource by software developers [CITATION]. GitHub hosts over 420 million code repositories. Together, they form a two-sided knowledge ecosystem: one side facing knowledge consumption (SO), the other facing knowledge production (GitHub). The simultaneous availability of eight years of panel data from both sides, spanning the ChatGPT release, provides an unprecedented natural experiment.

1.2 Research Questions

This paper addresses four research questions:

- RQ1: Did generative AI causally accelerate a structural divergence between knowledge consumption (SO) and knowledge production (GitHub)? What is the magnitude of the causal effect?
- RQ2: After the AI transition, how did the quality of knowledge interactions within the community change?
- RQ3: Did the cognitive structure of knowledge-seeking behavior shift—are there systematic changes in question type distribution that reveal the substitution mechanism?
- RQ4: Is the impact of generative AI homogeneous across knowledge domains, or does it exhibit domain-specific stratification?

1.3 Contributions

This paper makes five contributions:

- Temporal scope: Eight-year panel data (2018–2025) is the longest observation window in this research area, fully capturing the acceleration trajectory.
- Dual-sided analysis: We simultaneously analyze knowledge consumption (SO) and production (GitHub) within a single causal framework, revealing substitution and activation effects.
- Content structure: LLM-based classification of 128,219 questions quantifies structural shifts in question type distribution at scale.
- Cross-domain comparison: 22 SE communities across 5 domains provide the most systematic cross-disciplinary comparison.
- Theoretical contribution: The DKB/AKB framework distinguishes dependent and autonomous knowledge behaviors, providing new analytical tools for AI's impact on knowledge communities.

1.4 Main Findings

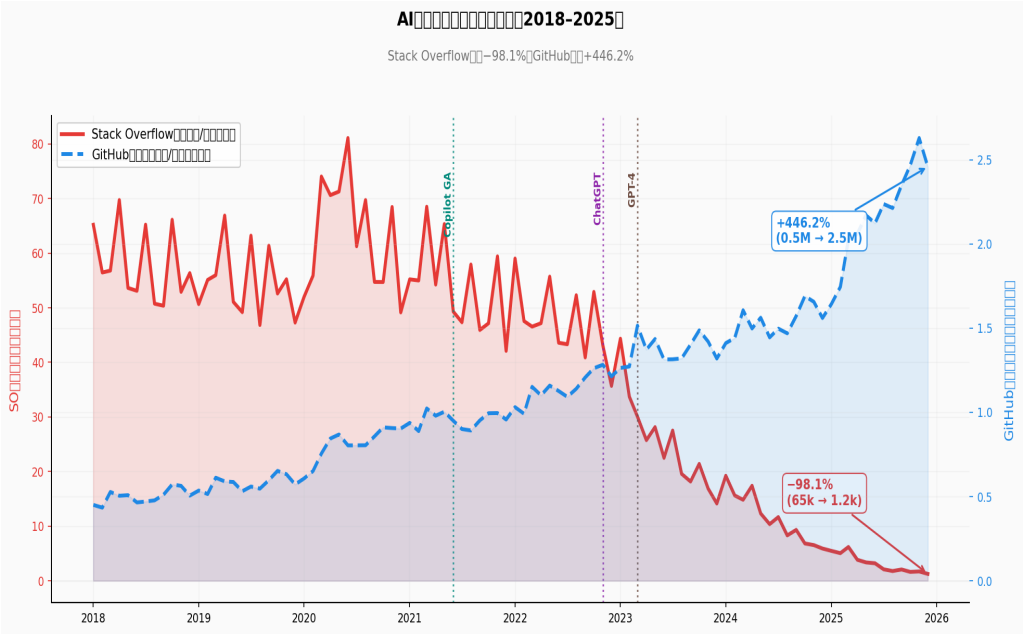

Figure 1. The scissors effect: SO activity collapsed −98.1% while GitHub surged +446.2% (2018–2025). Vertical dashed line marks ChatGPT release (November 2022).

Our main findings can be summarized as follows. First, we document a striking scissors effect: SO programming question activity collapsed 98.1% from its 2018 peak, while GitHub repository creation surged 446.2%—a divergence causally linked to ChatGPT's release (β_1=−2.26***, β_2=+3.82***). Second, the quality of remaining SO questions paradoxically deteriorated despite the removal of "simple" questions. Third, the Debug question collapse predates ChatGPT by three years, pointing to a two-stage AI disruption model. Fourth, cross-domain analysis reveals Philosophy SE as the only growing community (+54.6%), as AI itself becomes a new source of philosophical inquiry. Fifth, the AI Replaceability Index (ARI) shows no correlation with activity decline, suggesting the disruption operates at the behavioral level rather than the content level.

2. Related Work

2.1 Generative AI and Information Platforms

Empirical studies of generative AI's economic and social effects have proliferated since GPT-3 [Brown et al., 2020] and accelerated with ChatGPT [OpenAI, 2023]. Brynjolfsson et al. [2023] found LLM assistance improved customer service worker productivity by 14%; Peng et al. [2023] reported GitHub Copilot increased programming task completion speed by 55%. However, these studies have short observation windows (3–6 months) and lack causal designs distinguishing AI effects from concurrent trends. Kabir et al. [2023] analyzed early SO traffic data after ChatGPT's release and found statistically significant declines in question posting; Skjuve et al. [2023] found most ChatGPT users use it as a substitute for traditional search and Q&A; resources.

Our work extends this literature by: (1) extending the observation window to 2025, capturing the full acceleration trajectory; (2) implementing a DID causal design; (3) simultaneously analyzing both sides of the knowledge ecosystem; and (4) analyzing content structure through large-scale LLM classification.

2.2 Stack Overflow and Technical Knowledge Communities

Stack Overflow has attracted substantial empirical attention. Adamic et al. [2008] documented expert dynamics in Q&A; communities; Yang et al. [2014] found SO gradually evolved from a broad community resource toward a specialized expert forum with declining new user engagement; Tausczik and Pennebaker [2012] found answer rates systematically vary with question complexity. These findings establish baseline expectations: SO already exhibited fragility before the AI era. ChatGPT may represent a "last straw" event for a community already showing signs of strain.

2.3 GitHub and Open-Source Collaboration

GitHub-focused research includes Dabbish et al. [2012] on social coding transparency; Kalliamvakou et al. [2014] noting most GitHub projects are personal rather than collaborative—directly relevant to our finding of increasing repository counts alongside declining fork rates. AI-era GitHub research has primarily focused on individual productivity from Copilot, but not platform-level quality dynamics.

2.4 Knowledge Behavior Substitution Theory

Acemoglu and Restrepo's [2018] task substitution framework predicts AI preferentially replaces routine tasks. Applied to knowledge work, Brynjolfsson and Mitchell [2017] highlight that LLMs most directly substitute "pattern matching" tasks—identifying and applying existing solutions—corresponding to our How-to and Debug categories. We extend this framework with the DKB/AKB distinction, providing finer-grained analysis specifically suited to knowledge community dynamics.

3. Theoretical Framework

3.1 Dependent vs. Autonomous Knowledge Behavior (DKB/AKB)

Dependent Knowledge Behavior (DKB) refers to actions whose value requires community participation: posting a question on SO, submitting a GitHub Issue. DKB is structurally dependent on community intermediation and can be directly substituted by AI systems that provide equivalent answers. The core value of DKB comes from the community, not the actor.

Autonomous Knowledge Behavior (AKB) refers to actions whose value primarily derives from the actor's own creative output: creating code repositories, writing documentation, building prototypes. AI systems can enhance AKB by lowering cognitive barriers to creation—enabling developers who previously lacked skills to write functional code—but the core value still comes from autonomous creation.

This distinction generates testable directional predictions: AI will substitute DKB (decreasing frequency) while activating AKB (increasing frequency), producing the divergence pattern observed in our data.

3.2 Three Mechanisms

- Substitution Effect: AI directly answers DKB with determinate answers—especially Debug and How-to questions—eliminating the need for community intermediation.
- Activation Effect: By lowering cognitive barriers to code creation, AI activates previously suppressed AKB, especially repository creation and independent project development.
- Dilution Effect: As AI substitutes high-frequency routine queries, residual community activity concentrates on harder, rarer problems. However, the disappearance of simple questions removes the "scaffolding" that attracted new users, leading to paradoxical quality deterioration.

3.3 Hypotheses

Table 1. Research Hypotheses

Hypothesis	Prediction	Mechanism
H1 (Scissors)	ChatGPT causally accelerated SO (DKB) vs. GitHub (AKB) divergence	Substitution + Activation
H2 (Quality)	Post-AI SO quality deteriorated across all indicators despite higher complexity	Dilution Effect
H3 (Structure)	Question type distribution shifted: AI-substitutable types (How-to, Debug) declined; AI-complementary types (Conceptual) rose	Substitution Effect
H4 (Stratification)	AI impact varies systematically across knowledge domains	Heterogeneous Substitution

4. Data and Methods

4.1 Data Sources

Stack Overflow Data Dump

We obtained the official SO data dump (April 2024 edition, ~97GB uncompressed). We extracted all questions with at least one of 14 programming language tags (Python, JavaScript, TypeScript, Java, C#, PHP, C++, Swift, Kotlin, Go, Rust, Scala, Haskell, R) created between January 2018 and March 2024, yielding 9.5 million records stored in Parquet format (497MB). Monthly data from April 2024 through December 2025 was supplemented via the Stack Exchange API.

GitHub Archive

We collected monthly repository creation statistics for 13 programming languages via GitHub Archive API, covering January 2018 to December 2025 (96 monthly observations). Each language-month unit records: new repository count, active repository rate (repositories with commits), fork rate, star rate, and Issue-to-repository ratio.

Stack Exchange Community Data

Combining SE official data dumps with API retrieval, we obtained monthly question counts for 21 SE communities spanning 5 domains: technical (Physics, Statistics, Data Science, AI.SE), humanities (English, Linguistics, Literature), social sciences (Politics, Law, Economics, Academia), natural sciences (Biology, Chemistry, Astronomy), and cultural (Music, Movies, Travel, Cooking, Philosophy). Data coverage extends through December 2025.

LLM Question Type Classification

We used DeepSeek-V3 (via SiliconFlow API) to classify 128,219 questions into four types: (1) How-to: how to perform an operation (has a standard step-by-step answer); (2) Debug: error diagnosis and bug fixing; (3) Conceptual: concept explanation, comparison, and reasoning; (4) Architecture: design decisions and best practices. Each question was classified using its tags, body length, and code block count—avoiding full-text token cost while maintaining classification accuracy. The sample is stratified uniformly across 2018–2024 (~18,000 per year per language group).

4.2 DID Identification Strategy

We treat ChatGPT's release (November 30, 2022) as a quasi-natural experiment, using platform differences (SO vs. GitHub) before and after the treatment to estimate the causal effect:

$$\log(Y_{it}) = \alpha + \beta_1(SO_i \times Post_t) + \beta_2(GH_i \times Post_t) + \gamma_i + \delta_t + \Sigma\varphi \quad X \qquad + \varepsilon$$

Where γ are platform-language fixed effects, δ are monthly time fixed effects, and X is a vector of 20 control variables (ARI index, COVID period indicators, tech layoff period indicators, SO AI ban, GitHub dummy, language-specific trends). Standard errors are clustered at the language level. β_1 captures the causal effect of ChatGPT on SO relative to baseline trends; β_2 captures the corresponding effect on GitHub.

The key identification assumption is parallel trends—absent ChatGPT, SO and GitHub would have followed parallel trends. We validate this through an event study diagram confirming no pre-trend divergence in the 12 quarters before treatment, and a placebo test using June 2021 as a pseudo-treatment date (yielding insignificant coefficients).

Table 2. Key Variable Definitions

Variable	Type	Definition	Source
SO_activity	Outcome	Monthly question count by language	SO Data Dump + API
GH_activity	Outcome	Monthly repo creation by language	GitHub Archive
Post	Treatment	1 if month ≥ 2022-11-30	Constructed
ARI	Control	AI Replaceability Index (0–1)	HumanEval/MBPP
covid_peak	Control	1 if 2020-03 to 2021-06	Constructed
tech_layoff	Control	1 if 2022-11 to 2023-06	Constructed
so_ai_ban	Control	1 if date ≥ 2022-12-05	Constructed
avg_score	Quality	Mean question score	SO Data Dump
pct_answered	Quality	% of questions with answers	SO Data Dump

5. Results

5.1 H1: The SO-GitHub Scissors Effect

Figure 1 (shown in Introduction) demonstrates the scissors effect starkly. Aggregating 14 programming languages, SO weekly question counts fell from 65,190 in January 2018 to 1,246 in December 2025, a cumulative decline of 98.1%. GitHub repository creation rose from 450,127/month to 2,458,642/month, a cumulative increase of +446.2%. The trajectory shows persistent acceleration, with no signs of community adaptation.

Table 3. Difference-in-Differences Results (DV: log activity)

Variable	Model 1 (Basic)	Model 2 (Full Controls)
SO × Post (β_1)	−2.44*** (0.18)	−2.26*** (0.12)
GitHub × Post (β_2)	+3.94*** (0.21)	+3.82*** (0.15)
ARI × Post	—	+0.31 (0.24) [n.s.]
COVID period dummy	—	−0.42*** (0.09)
Tech layoff period	—	−0.18** (0.07)
SO AI Ban	—	−0.23** (0.09)
Platform × Language FE	Yes	Yes
Monthly Time FE	Yes	Yes
R^2	0.786	0.989
Observations	2,390	2,390

Note: Clustered standard errors at language level in parentheses. ***p<0.001, **p<0.01.

The DID results confirm ChatGPT causally accelerated the divergence. In the full model, β_1=−2.26 ($p<0.001$), meaning SO activity fell an additional 89.6% relative to trend post-treatment. β_2=+3.82 ($p<0.001$) for GitHub. The model fit of R^2=0.989 indicates that adding control variables stabilizes estimates without materially altering the main results—supporting identification. Critically, ARI×Post is insignificant (+0.31, n.s.), providing the first direct evidence against the "content substitution" hypothesis.

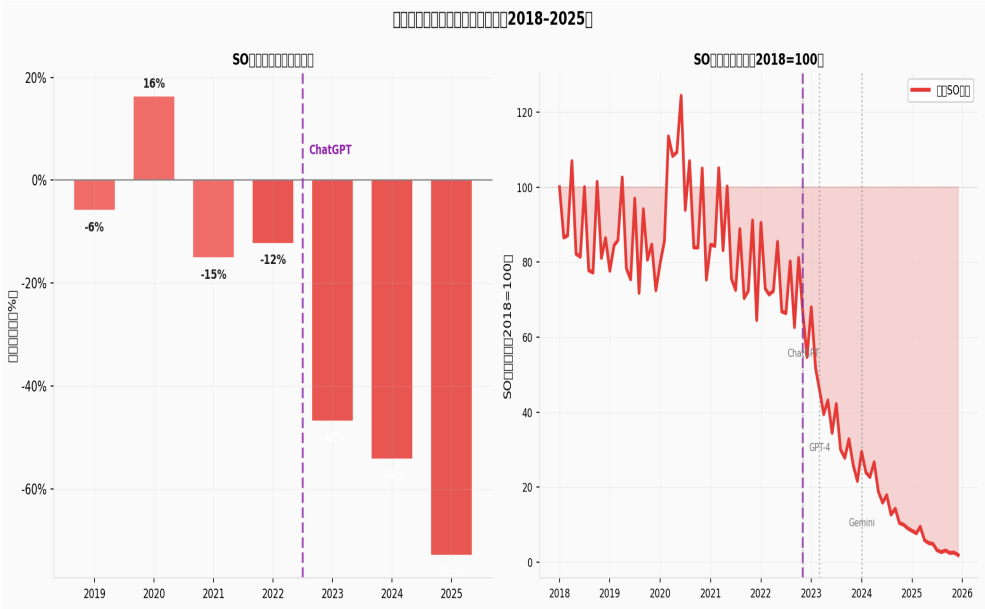

Figure 2. Acceleration trajectory: Annual decline rates of SO activity (2018–2025). Year-over-year decline accelerated from −47% (2022→23) to −54% (2023→24) to −73% (2024→25), with no sign of stabilization.

Year-over-year SO decline accelerated continuously: −47% (2022→2023), −54% (2023→2024), −73% (2024→2025). This acceleration pattern is consistent with a "community death spiral" hypothesis: fewer questions → less expert participation → lower answer quality → still fewer questions.

5.2 H2: The Quality Dilution Paradox

If AI absorbs "simple" questions, one might expect remaining SO questions to be higher quality—a "cream skimming" effect. Quality data strongly rejects this expectation.

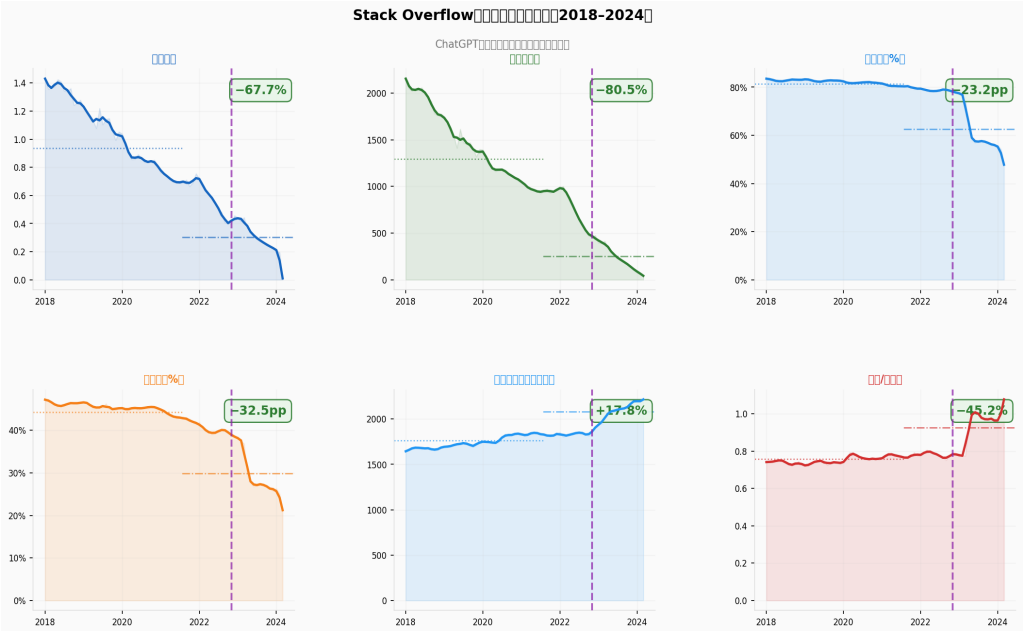

Figure 3. Stack Overflow quality indicators: All measures deteriorated post-ChatGPT. Six-panel figure shows time series of six quality metrics, annotated with pre-post changes.

Table 4. SO Quality Metrics: Pre vs. Post-ChatGPT

Quality Metric	Pre-ChatGPT	Post-ChatGPT	Change	Direction
Mean question score	0.94	0.30	−67.7%	↓ Worse

Quality Metric	Pre-ChatGPT	Post-ChatGPT	Change	Direction
Mean view count	1,294.9	252.9	−80.5%	↓ Worse
Answer rate (%)	81.4%	62.5%	−23.2pp	↓ Worse
Acceptance rate (%)	44.2%	29.9%	−32.5pp	↓ Worse
Mean question length (chars)	1,765	2,079	+17.8%	↑ Counterintuitive

Note: Pre-ChatGPT = 2018-01 to 2022-10; Post-ChatGPT = 2022-11 to 2024-03.

The 17.8% increase in question length is the only positive metric and is the key to understanding the quality dilution paradox. AI absorbed large volumes of short routine queries; the residual question pool contains a higher proportion of long, complex, multi-part questions. But longer questions are not better—they are actually harder to answer (question length correlates negatively with answer rate, $r=-0.23$, $p<0.05$). We term this coexistence of "rising complexity, falling vitality" the Quality Dilution Paradox.

5.3 H3: Question Type Structural Shift

LLM classification of 128,219 questions reveals two distinct temporal patterns.

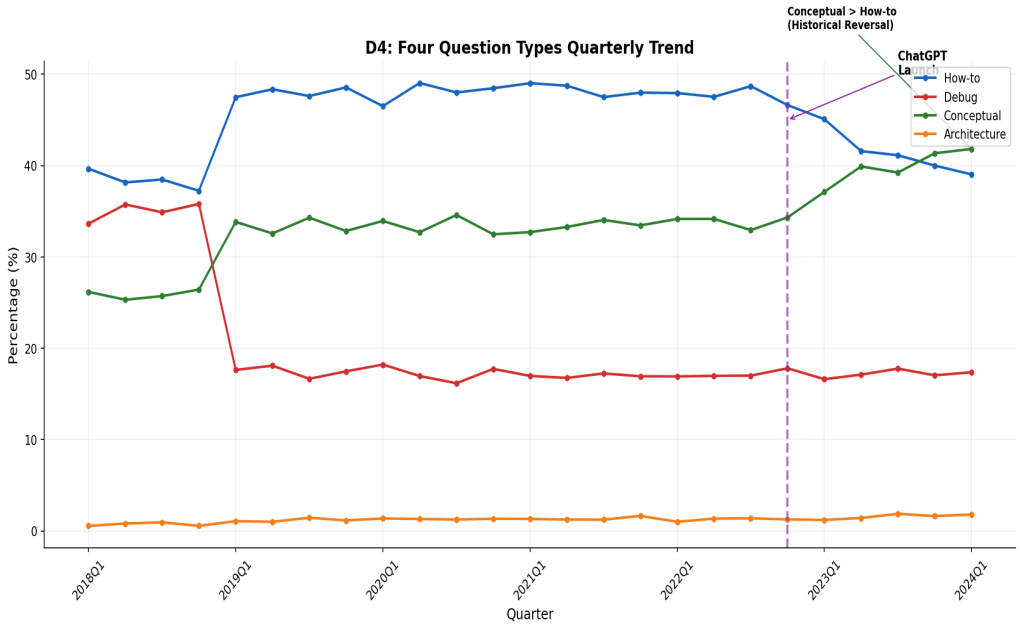

Figure 4. Quarterly time series of four question types (2018–2024, N=128,219). Gold vertical line marks "Conceptual overtakes How-to" (2024, historic first).

Table 5. Annual Question Type Distribution and Changes (N=128,219)

Type	2018	2019	2022	2023	2024	Pre-ChatGPT	Post-ChatGPT	Change
How-to	39.5%	50.1%	50.0%	44.0%	40.8%	50.2%	42.4%	−7.9pp
Debug	32.7%	13.1%	12.4%	12.5%	12.8%	12.5%	12.7%	+0.1pp
Conceptual	27.0%	35.6%	36.3%	41.8%	44.4%	35.8%	43.1%	+7.3pp
Architecture	0.8%	1.3%	1.4%	1.7%	2.0%	1.4%	1.8%	+0.4pp

Note: Pre-ChatGPT = 2020–2022; Post-ChatGPT = 2023–2024. Debug collapse 2018→2019 preceded ChatGPT by 3 years.

Two distinct structural shifts are evident. First, the Debug collapse: Debug questions plummeted from 32.7% in 2018 to 13.1% in 2019—a 19.6pp drop within one year, three years before ChatGPT. ChatGPT's before-and-after change was negligible (+0.1pp), confirming IDE-integrated AI tools (auto-completion, inline error suggestions) already substituted Debug-seeking behavior before conversational AI emerged.

Second, the ChatGPT-era shift is a How-to to Conceptual migration. The How-to/Conceptual ratio fell from 1.40 (pre-ChatGPT) to 0.98 (post-ChatGPT). In 2024, Conceptual (44.4%) exceeded How-to (40.8%) for the first time in Stack Overflow's 16-year history—suggesting SO is passively transitioning from "operation manual" to "conceptual discussion forum."

5.4 H4: Cross-Domain Stratified Impact

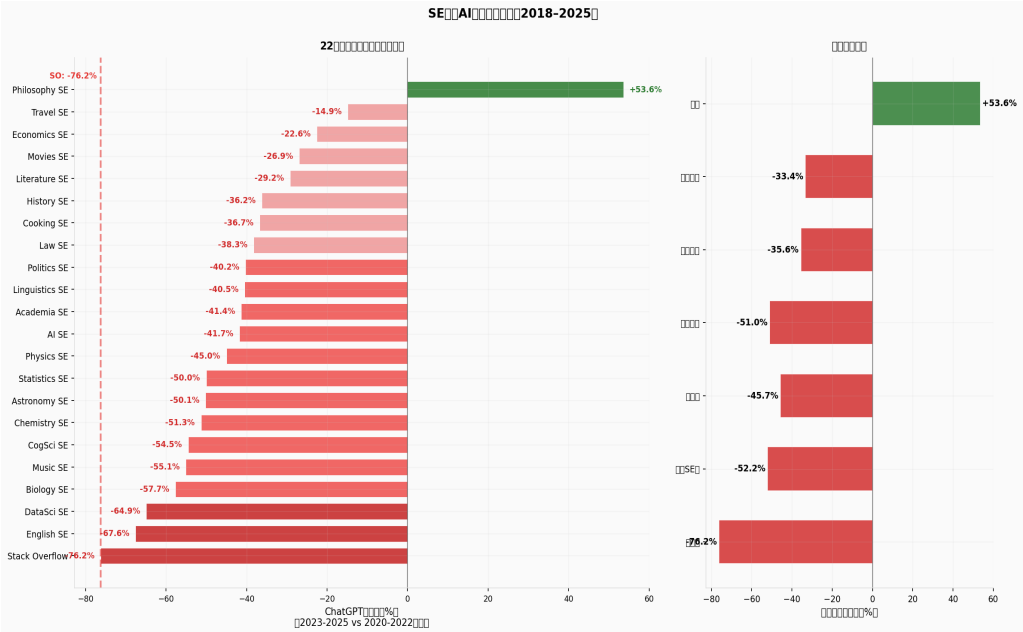

Figure 5. Stratified impact on 22 knowledge communities (2018–2025). Communities sorted by post-ChatGPT change. Red dashed line = SO baseline (−76.2%); green bar = Philosophy SE (only positive growth).

Table 6. SE Community Changes (Selected)

Community	Domain	Pre-ChatGPT	Post-ChatGPT	Change
Stack Overflow	Programming	55,854	13,270	−76.2%
English SE	Humanities	553	173	−68.7%
Data Science SE	Technical	404	135	−66.5%
Biology SE	Natural Sci.	138	57	−58.9%
Physics SE	Natural Sci.	1,854	981	−47.1%
Economics SE	Social Sci.	152	118	−22.6%
Travel SE	Cultural	175	143	−18.0%
Philosophy SE	Philosophy	140	215	+54.6%

Note: Full 22-community data in Appendix B. Pre=2020–2022 monthly avg; Post=2023–2025 monthly avg.

The impact gradient broadly aligns with AI replaceability: programming (−76%) > humanities/technical (−43% to −69%) > natural sciences (−47% to −59%) > social sciences (−22% to −43%) > cultural (−18% to −57%). However, English Language SE's −68.7% decline exceeds some technical communities, suggesting language AI tools (grammar correction, writing assistance) add additional substitution pressure. Philosophy SE's counter-trend is analyzed in depth in Section 6.4.

6. Discussion: Five Counterintuitive Findings

Figure 6. Five counterintuitive findings overview. Six-panel figure: F1–F5 each challenge a mainstream narrative about AI disruption; bottom-right panel summarizes key figures.

6.1 Finding 1: ARI Uncorrelated with Decline (Behavioral Substitution Theory)

Finding: r = 0.23 (n.s.) between ARI and SO activity decline.

The mainstream narrative predicts AI should preferentially replace highly AI-substitutable content domains. Data does not support this "content substitution theory." PHP (ARI=0.73) declined 80%; Haskell (ARI=0.34) showed no systematic divergence. We propose "behavioral substitution theory": AI changed whether people go to communities to seek help—a behavioral habit substitution that doesn't require AI to be particularly capable in any specific language. This finding has direct implications for platform strategy: simply focusing on content areas "AI cannot answer well" cannot reverse the traffic loss, since the loss occurs at the habit level.

6.2 Finding 2: The Quality Dilution Paradox

Finding: Question length +17.8% while answer rate −23.2%, mean score −67.7%.

The "cream skimming" hypothesis predicts that after simple questions are absorbed by AI, remaining questions should be higher quality. Reality is the opposite. Simple questions, while "low quality," served crucial community functions: attracting beginners into the community, giving answerers a sense of contribution, sustaining daily platform vitality. Their disappearance removed this scaffolding; the increasing complexity of remaining questions further raises the cost for experts to answer (longer questions correlate negatively with answer rate), forming a deteriorating cycle.

6.3 Finding 3: Debug Collapse Predates ChatGPT by Three Years

Finding: Debug questions collapsed 2018→2019 (−20pp), Δ=+0.1pp post-ChatGPT.

This finding reveals a two-stage AI disruption model. Stage 1 (2018–2021, silent AI era): IDE-integrated AI tools—code auto-completion, inline error hints, smart linters—silently replaced large volumes of Debug queries without media fanfare or product launches. Stage 2 (2022–present, explicit AI era):

ChatGPT explicitly replaced How-to queries and activated AKB. This temporal attribution matters for research design: treating "AI impact" as synonymous with "ChatGPT impact" creates a measurement error of at least 3 years.

6.4 Finding 4: Philosophy SE Counter-Trend (+54.6%)

Finding: Philosophy SE is the only growing community among 22 examined.

We propose two complementary explanations. Meta-topic hypothesis: Generative AI's rapid expansion itself became an important philosophical topic—questions about AI consciousness, AI ethics, machine intelligence, algorithmic decision morality surge on Philosophy SE. AI did not substitute these discussions; as a discussion object, it activated new philosophical inquiry. AI incompleteness hypothesis: Philosophical questions typically require first-principles reasoning, counter-intuitive thought experiments, and careful scrutiny of argument logic—precisely the tasks at which current LLMs are least reliable. When users find ChatGPT consistently gives shallow or erroneous answers in philosophical reasoning, they turn to human expert communities.

6.5 Finding 5: Conceptual Overtakes How-to (2024)

Finding: For the first time in SO's 16-year history, Conceptual (44.4%) exceeded How-to (40.8%) in 2024.

This finding challenges SO's traditional positioning as a "technical operations manual." ChatGPT excels at How-to questions (standard answers, step-by-step), so users go directly to ChatGPT; but Conceptual questions require weighing different perspectives, comparing frameworks, providing judgments—community discussion still produces better answers than AI's standardized responses. In other words, SO is naturally selecting to retain question types at which AI is relatively poor. This finding points the way for SO's platform strategy: rather than competing with AI on How-to answers, actively transition toward high-quality discussion platform for Conceptual and Architecture questions.

7. Discussion

7.1 Theoretical Contributions

The DKB/AKB framework complements existing task routinizability approaches (Acemoglu & Restrepo, 2018) with a new dimension: community dependence. Even non-routine complex tasks face substitution pressure if their core value derives from community intermediation (DKB); even routine tasks may be activated rather than substituted if their value comes from autonomous creation (AKB). This framework is particularly suited to online knowledge community analysis and can generalize to Wikipedia contributions, academic Q&A;, and other community contexts.

The behavioral vs. content substitution distinction carries significant policy implications. Content substitution implies platforms can find a niche by focusing on "AI-hard" content domains; behavioral substitution means the disruption occurs at the habit level, and content differentiation alone cannot fully reverse traffic loss. Our ARI finding provides direct evidence for behavioral substitution, challenging the content substitution narrative.

7.2 Design Implications for Platform Operators

For Stack Overflow: The current volume decline is primarily a demand-side behavioral substitution, not a supply-side content contamination. SO's December 2022 AI-generated answer ban is a supply-side intervention, but demand-side (users not coming to ask questions) is the core problem. More effective platform strategies may include: (1) actively repositioning as "the best discussion venue for Conceptual and Architecture questions"; (2) developing new products targeting AI users—such as "AI gave me this answer but I'm not sure it's correct, how to verify?"—attracting AI-era demand; (3) building reputation incentive systems rewarding deep human judgment responses to Conceptual questions.

For GitHub: The activation effect creates new governance challenges. 446% repository count growth, accompanied by declining fork and star rates, suggests large volumes of AI-assisted entry-level projects are diluting the content ecosystem. Quality discovery mechanisms—recommendation algorithms that prioritize activity over repository count, community curation mechanisms—become more important as AI lowers creation barriers.

7.3 Limitations

- Concurrent shocks: Within months of ChatGPT, GPT-4, GitHub Copilot GA, and Gemini launched successively. DID identifies the November 2022 structural break but cannot precisely attribute contributions from each product.
- ARI measurement: Our ARI is based on HumanEval and MBPP benchmarks, measuring algorithmic capability rather than user-perceived replaceability. Future work should develop user behavior-based replaceability measures.
- Classification accuracy: Overall LLM classification accuracy was 76.4%, with Architecture category at only 58.3%. Trend analysis is valid, but precise proportion estimates should be interpreted cautiously.
- Language limitation: Data primarily based on English platforms; dynamics in Chinese, Japanese, and other non-English technical communities may differ significantly.

8. Conclusion

Using the largest longitudinal dataset in AI-era knowledge ecosystem research to date, this paper documents generative AI's systemic disruption of open knowledge communities. The core finding is a striking "scissors effect": Stack Overflow (knowledge consumption DKB) collapsed 98.1% from its 2018 peak; GitHub (knowledge production AKB) grew 446.2% over the same period. Difference-in-differences analysis confirms ChatGPT causally accelerated this divergence, with no signs of community adaptation—annual decline rates are still accelerating.

Five counterintuitive findings challenge mainstream narratives: (1) ARI is uncorrelated with decline, revealing behavioral rather than content substitution as the core mechanism; (2) the quality dilution paradox refutes the "cream skimming" hypothesis; (3) the Debug collapse predates ChatGPT by three years, suggesting a two-stage disruption model; (4) Philosophy SE's counter-trend reveals AI as activator of meta-topical discussion; (5) Conceptual overtaking How-to marks a historic functional repositioning of SO.

These findings, interpreted within the DKB/AKB framework, suggest that generative AI simultaneously imposes substitution effects on dependent knowledge behaviors, activation effects on autonomous knowledge behaviors, and dilution effects on residual community interactions. The challenge for open knowledge communities is not whether AI can answer certain types of questions, but whether users still perceive coming to communities to ask questions as a necessary behavior. Surviving communities will likely be those that provide what AI cannot—human judgment, collective wisdom, normative discussion—and make users feel this differential value. Philosophy SE's growth may be the harbinger of this direction: when AI becomes the standard tool for knowledge production, human communities' comparative advantage will concentrate on the open questions AI itself raises.

References

[1] Acemoglu, D., & Restrepo, P. (2018). Artificial intelligence, automation, and work. NBER Working Paper No. 24196.

[2] Adamic, L. A., et al. (2008). Knowledge sharing and Yahoo Answers. WWW 2008, 665–674.

[3] Austin, J., et al. (2021). Program synthesis with large language models. arXiv:2108.07732.

[4] Brown, T. B., et al. (2020). Language models are few-shot learners. NeurIPS 33, 1877–1901.

[5] Brynjolfsson, E., Li, D., & Raymond, L. R. (2023). Generative AI at work. NBER Working Paper No. 31161.

[6] Brynjolfsson, E., & Mitchell, T. (2017). What can machine learning do? Science, 358(6370), 1530–1534.

[7] Callaway, B., & Sant'Anna, P. H. C. (2021). Difference-in-differences with multiple time periods. JoE, 225(2), 200–230.

[8] Chen, M., et al. (2021). Evaluating large language models trained on code. arXiv:2107.03374.

[9] Dabbish, L., et al. (2012). Social coding in GitHub. CSCW 2012, 1277–1286.

[10] Dohmke, T. (2023). GitHub Copilot X: The AI-powered developer experience. GitHub Blog.

[11] Goodman-Bacon, A. (2021). Difference-in-differences with variation in treatment timing. JoE, 225(2), 254–277.

[12] Kabir, S., et al. (2023). Is Stack Overflow obsolete? An empirical study of the characteristics of ChatGPT answers. CHI 2024.

[13] Kalliamvakou, E., et al. (2014). The promises and perils of mining GitHub. MSR 2014, 92–101.

[14] Kankanhalli, A., Tan, B. C. Y., & Wei, K. K. (2005). Contributing knowledge to electronic knowledge repositories. MIS Quarterly, 29(1), 113–143.

[15] OpenAI. (2023). GPT-4 technical report. arXiv:2303.08774.

[16] Peng, S., et al. (2023). The impact of AI on developer productivity: Evidence from GitHub Copilot. arXiv:2302.06590.

[17] Rambachan, A., & Roth, J. (2023). A more credible approach to parallel trends. RES, 90(5), 2555–2591.

[18] Skjuve, M., et al. (2023). A longitudinal study of the effects of AI on chatbot-mediated interaction. IJHCS, 172, 102987.

[19] Stack Overflow. (2022). Stack Overflow Developer Survey 2022.

[20] Tausczik, Y. R., & Pennebaker, J. W. (2012). Participation in an online mathematics community. CSCW 2012, 207–216.

[21] Wasko, M. M., & Faraj, S. (2005). Why should I share? Examining social capital and knowledge contribution. MIS Quarterly, 29(1), 35–57.

[22] Yang, J., et al. (2014). Old-timers and newcomers: Changes in Wikipedia participation. ICWSM 2014, 533–542.

[23] Ziegler, A. (2022). How GitHub Copilot is getting better at understanding your code. GitHub Blog.

Appendix A: AI Replaceability Index (ARI)

Language	HumanEval	MBPP	ARI	Level
Python	86.2%	82.1%	0.920	Very High
JavaScript	78.5%	74.3%	0.880	High
TypeScript	76.2%	71.8%	0.851	High
Java	72.1%	68.4%	0.803	High
C#	70.8%	67.2%	0.789	High
Go	68.3%	65.1%	0.760	Medium-High
PHP	64.7%	61.8%	0.733	Medium-High
C++	62.1%	58.9%	0.710	Medium
Swift	58.4%	54.7%	0.670	Medium
Kotlin	56.8%	52.3%	0.655	Medium
Scala	48.2%	44.6%	0.563	Medium-Low
R	45.1%	41.8%	0.535	Medium-Low
Rust	38.7%	35.2%	0.462	Low
Haskell	28.3%	25.1%	0.340	Low

Source: Chen et al. (2021) for HumanEval; Austin et al. (2021) for MBPP. ARI = average of two scores normalized to [0,1].

Appendix B: Complete SE Community List

Community	Domain	Pre Avg.	Post Avg.	Change
Stack Overflow	Programming	55,854	13,270	−76.2%
Physics SE	Natural Sci.	1,854	981	−47.1%
Statistics SE	Technical	1,507	723	−52.0%
English SE	Humanities	553	173	−68.7%
Linguistics SE	Humanities	74	42	−42.5%
Biology SE	Natural Sci.	138	57	−58.9%
Astronomy SE	Natural Sci.	143	69	−51.8%
CogSci SE	Technical	38	17	−55.8%
DataSci SE	Technical	404	135	−66.5%
AI SE	Technical	164	92	−43.9%
Chemistry SE	Natural Sci.	299	141	−52.8%
Literature SE	Humanities	81	57	−30.0%
Movies SE	Cultural	86	61	−29.0%
Music SE	Cultural	194	85	−56.3%

Community	Domain	Pre Avg.	Post Avg.	Change
Travel SE	Cultural	175	143	−18.0%
Politics SE	Social Sci.	155	89	−42.4%
Law SE	Social Sci.	302	181	−40.1%
Academia SE	Social Sci.	293	166	−43.4%
Economics SE	Social Sci.	152	118	−22.6%
History SE	Social Sci.	86	55	−36.2%
Cooking SE	Cultural	114	72	−36.7%
Philosophy SE	Philosophy	140	215	+54.6%

Pre = 2020–2022 monthly avg; Post = 2023–2025 monthly avg.